

Phase 0 - Project Definition

- ☐ Define the business problem
- ☐ Define project scope
- ☐ Define out-of-scope items
- ☐ Define target users
- ☐ Define business questions the platform should answer
- ☐ Define success criteria
- ☐ Select primary FAOSTAT datasets/domains
- ☐ Decide initial historical period
- ☐ Create project README
- ☐ Create project glossary

Deliverable: Project Charter

Phase 1 - Local Development & Git Setup

- ☐ Create local project directory
- ☐ Initialize Git repository
- ☐ Create GitHub repository
- ☐ Create production-style folder structure
- ☐ Create `.gitignore`
- ☐ Create Python virtual environment
- ☐ Create dependency file
- ☐ Configure VS Code
- ☐ Define branch strategy
- ☐ Define commit message convention
- ☐ Add initial README
- ☐ Push initial repository

Deliverable: Clean development workspace and GitHub repository

Phase 2 - Data Discovery & Source Profiling

FAOSTAT

- ☐ Study FAOSTAT API documentation
- ☐ Identify relevant domains
- ☐ Understand dimensions
- ☐ Understand indicator definitions
- ☐ Understand country/area codes
- ☐ Understand year granularity
- ☐ Investigate missing values
- ☐ Investigate revisions

- ☐ Inspect API pagination
- ☐ Inspect API response formats
- ☐ Pull a small sample locally
- ☐ Profile sample JSON/CSV
- ☐ Document source quirks

World Bank

- ☐ Identify possible enrichment indicators
- ☐ Understand World Bank country identifiers
- ☐ Investigate mapping with FAOSTAT countries
- ☐ Document potential integration strategy

Deliverable: Source Analysis Document + Sample Data

Phase 3 - Architecture & Data Contracts

- ☐ Draw logical architecture
- ☐ Draw physical architecture
- ☐ Define source-to-target flow
- ☐ Define RAW responsibilities
- ☐ Define CLEAN responsibilities
- ☐ Define CONSUMPTION responsibilities
- ☐ Define PUBLISH responsibilities
- ☐ Define ingestion metadata
- ☐ Define naming conventions
- ☐ Define schema conventions
- ☐ Define data types
- ☐ Define data contracts
- ☐ Define expected refresh frequency
- ☐ Define retry behaviour
- ☐ Define duplicate-handling strategy
- ☐ Define schema-evolution strategy
- ☐ Create ADR-001 for major architecture decisions

Deliverable: Architecture v1

Phase 4 - Snowflake Platform Foundation

- ☐ Design Snowflake databases
- ☐ Design schemas
- ☐ Design warehouses
- ☐ Define warehouse sizing assumptions
- ☐ Configure auto-suspend
- ☐ Configure auto-resume

- [] Define Snowflake roles
- [] Define service roles
- [] Define developer role
- [] Define analyst/consumer role
- [] Create base Snowflake objects
- [] Validate permissions

Deliverable: Snowflake platform skeleton

Phase 5 - AWS Landing Zone

- [] Create AWS S3 bucket
- [] Design bucket prefix structure
- [] Define raw-file naming convention
- [] Understand IAM users vs roles
- [] Create Snowflake-compatible IAM permissions
- [] Configure Snowflake Storage Integration
- [] Create external stage
- [] Validate Snowflake → S3 access
- [] Define retention/lifecycle approach
- [] Ensure credentials are never committed to Git

Example logical structure:

```
source/domain/year/month/day/run_id/file
```

Deliverable: Secure AWS S3 → Snowflake connection

Phase 6 - Source Ingestion

- [] Build FAOSTAT extraction module
- [] Handle pagination
- [] Handle HTTP errors
- [] Handle retries
- [] Handle timeouts
- [] Capture request metadata
- [] Generate ingestion/run ID
- [] Capture extraction timestamp
- [] Write source files without modifying them
- [] Upload files to S3
- [] Make ingestion idempotent
- [] Log row/file counts
- [] Test partial failures
- [] Test repeated executions

Deliverable: Re-runnable FAOSTAT → S3 ingestion pipeline

Phase 7 - RAW / Landing Layer

- ☐ Create appropriate file formats
- ☐ Create RAW tables
- ☐ Load files from S3
- ☐ Preserve original source attributes
- ☐ Capture filename
- ☐ Capture ingestion timestamp
- ☐ Capture run ID
- ☐ Capture source-system metadata
- ☐ Prevent duplicate loads
- ☐ Create load auditing
- ☐ Validate source vs target counts

Deliverable: Auditable immutable RAW layer

Phase 8 - CLEAN / Standardization Layer

- ☐ Parse source structures
- ☐ Flatten nested data where required
- ☐ Normalize column names
- ☐ Standardize country identifiers
- ☐ Standardize indicator identifiers
- ☐ Standardize years/dates
- ☐ Standardize units
- ☐ Handle null values
- ☐ Handle duplicates
- ☐ Handle malformed records
- ☐ Implement data-quality rules
- ☐ Separate rejected records where appropriate
- ☐ Document transformation rules

Deliverable: Trusted standardized datasets

Phase 9 - World Bank Enrichment

- ☐ Select useful economic/demographic indicators
- ☐ Build World Bank API ingestion
- ☐ Land World Bank raw data in S3
- ☐ Load RAW data into Snowflake
- ☐ Standardize World Bank data

- ☐ Build country crosswalk
- ☐ Handle unmatched countries
- ☐ Integrate World Bank and FAOSTAT data

Deliverable: Multi-source integrated dataset

Phase 10 - Dimensional Modeling

- ☐ Identify business grain
- ☐ Design fact tables
- ☐ Design country dimension
- ☐ Design indicator dimension
- ☐ Design date/year dimension
- ☐ Design geography/region dimension
- ☐ Consider source dimension
- ☐ Determine surrogate-key strategy
- ☐ Determine SCD requirements
- ☐ Define relationships
- ☐ Validate model against business questions
- ☐ Create ER/data-model diagram

Deliverable: Consumption-layer dimensional model

Phase 11 - Data Quality Framework

- ☐ Test uniqueness
- ☐ Test mandatory fields
- ☐ Test accepted values
- ☐ Test relationships
- ☐ Test year ranges
- ☐ Test numerical ranges
- ☐ Test duplicate detection
- ☐ Test referential integrity
- ☐ Validate source-vs-target counts
- ☐ Track rejected records
- ☐ Store quality-test results
- ☐ Decide pipeline behaviour on failed checks

Deliverable: Automated data-quality suite

Phase 12 - Transformation & Orchestration

- ☐ Understand Streams
- ☐ Understand Tasks

- ☐ Understand Dynamic Tables
- ☐ Compare appropriate approaches
- ☐ Select orchestration design
- ☐ Implement incremental processing
- ☐ Configure dependencies
- ☐ Configure schedules
- ☐ Configure retries where applicable
- ☐ Test reruns
- ☐ Test missed schedules
- ☐ Test duplicate source arrival

Deliverable: Automated Snowflake processing pipeline

Phase 13 - Security & Governance

- ☐ Implement RBAC
- ☐ Apply least-privilege access
- ☐ Separate service and human roles
- ☐ Restrict RAW access
- ☐ Review ownership
- ☐ Review future sensitive-data handling
- ☐ Configure appropriate grants
- ☐ Document access model
- ☐ Test unauthorized-access scenarios

Deliverable: Production-style security model

Phase 14 - Monitoring & Cost Management

- ☐ Monitor warehouse usage
- ☐ Monitor query history
- ☐ Monitor pipeline runs
- ☐ Monitor failed loads
- ☐ Monitor data freshness
- ☐ Monitor data-quality failures
- ☐ Track processed records
- ☐ Track processing duration
- ☐ Examine Snowflake credit usage
- ☐ Tune warehouse configuration
- ☐ Detect expensive queries
- ☐ Create operational queries/dashboard

Deliverable: Observability + cost-control layer

Phase 15 - Consumption & Streamlit

- ☐ Define dashboard personas
- ☐ Define KPIs
- ☐ Build analytical views
- ☐ Build food-security trends
- ☐ Build country comparisons
- ☐ Build regional comparisons
- ☐ Add economic/demographic enrichment
- ☐ Build Streamlit application
- ☐ Add filters
- ☐ Validate calculations
- ☐ Review query performance

Deliverable: Global Food Security Intelligence application

Phase 16 - CI/CD

- ☐ Create GitHub Actions workflow
- ☐ Protect credentials with GitHub Secrets
- ☐ Add Python checks
- ☐ Add SQL validation
- ☐ Add automated tests
- ☐ Create development deployment workflow
- ☐ Define promotion strategy
- ☐ Understand DEV → TEST → PROD deployment concepts
- ☐ Prevent accidental production deployment

Deliverable: Repeatable deployment process

Phase 17 - Production Failure Drills

Intentionally break things.

- ☐ Duplicate source file
- ☐ Missing file
- ☐ Bad JSON/CSV
- ☐ API timeout
- ☐ HTTP failure
- ☐ New source column
- ☐ Missing source column
- ☐ Invalid country code
- ☐ Pipeline interruption halfway through
- ☐ Rerun historical period

- ☐ Incorrect transformation
- ☐ Warehouse accidentally left running
- ☐ Unauthorized-user test

For each incident:

- ☐ Detect
- ☐ Diagnose
- ☐ Recover
- ☐ Prevent recurrence
- ☐ Document resolution

Deliverable: Incident/Recovery Runbook

Phase 18 - Final Production Review

- ☐ Review architecture
- ☐ Review repository
- ☐ Review SQL quality
- ☐ Review Python quality
- ☐ Review security
- ☐ Review costs
- ☐ Review scalability
- ☐ Review idempotency
- ☐ Review observability
- ☐ Review documentation
- ☐ Review business usefulness
- ☐ Identify technical debt
- ☐ Produce architecture diagram v2
- ☐ Complete final README
- ☐ Produce project retrospective

Final Deliverable: Production-style Snowflake + AWS data engineering portfolio project